

be able to communicate with each other. In practice though, interconnecting processors can prove to be prohibitively expensive; various interconnect structures have been proposed and adopted. They vary from crossbar connections to data bus-style connections. Again, the interconnect structures limit the number of processors, especially with an increase in processor speeds.

Distributed Memory MIMD (DM-MIMD): In this sort of memory organisation, each processor has its own memory. Processors are connected by a high-speed network switch that allows for data exchange between their respective memory locations. To use DM-MIMD, the user has to explicitly distribute data across various processors, and then arrange for data exchange between them—unlike the SM-MIMD architecture, which allows for transparent memory addressing across all processors. However, this is amongst the fastest growing class of HPC machines today, and with good reason—within current standards, this should outperform all others. DM-MIMD machines have no crippling memory bandwidth problems, or restrictions on the number of processors.

DM-MIMD machines also have their disadvantages—communications between processors are much slower than in SM-MIMD systems. In computations that require frequent data exchange and synchronisation between processors, performance may be below par.

Cache Coherent Non-Uniform Memory Access Machines (ccNUMA):

This forms a part of a growing trend in designing supercomputing machines. Systems built on this approach usually have a small number of processors tightly integrated or clustered into a Symmetric Multi-Processing (SMP) node. The processors within the nodes are connected by a swift crossbar interconnect, and those clusters are then connected by high speed network cabling. ccNUMA lets the SMP nodes share their memory—all the memory in the system is addressed globally. This means that a data item may not be located in physical memory, but may be within one logical shared address space—data is physically distributed over several nodes. Since the memory is globally addressed, it's important to keep consistency in the data values throughout—thus the name 'cache coherent'. All caches ensure that the values of variables held by them are coherent and consistent. Several supercomputers in the world are ccNUMA machines, such as the SGI Origin3000 series and the HP Superdome.



India's Big Iron, at CDAC's Bangalore supercomputing facility: Seen here (white and red), the CDAC-developed Param Padma supercomputing cluster

In front (black): A Sun UltraSPARC high performance computing machine

Super processors

Looking into the past, supercomputers used rather expensive proprietary processors; today's supercomputers use a wide variety of processors, from RISC processors such as those from HP's PA-RISC and Alpha families, to the better-known AMD Opteron, and the Intel Itanium and Pentium series.

RISC processors: RISC processors are inherently different from the more common-place CISC processors. They were, and still are, preferred—even with lower

Questions You Always Wanted Answered

How does a supercomputer bootstrap?

You kick it with your boot! The same as a normal computer, except that most of them are server-level machines. All you do is turn on the power.

Can I Quake on a supercomputer?

Only *Quake XI* at 6,500,000 fps! Seriously, no. High performance machines work only with UNIX or its clones, though HP SuperDome does run a variety of Windows. Now all you need is the requisite hardware and drivers, not to mention support for *Quake*.

I have two computers and an empty closet. Can I rig up a ccNUMA machine?

No! You need a lot more money, lethal hardware, about 2,000 sq ft of enclosed space with air-conditioning, several miles of cabling and some serious technical guidance.

My new Pentium 4 packs more punch than a CRAY XMP... I have super computing power, don't I?

Dream on! Comparing a decade old supercomputer to your 3-month old PC isn't quite right. Both work great, but work best for the applications they were designed for.

So I can't have my own supercomputer?

Sure you can, for Rs 30,00,00,000 only! What you can do is set up your own cluster, and simulate a cluster supercomputer.

Would that work?

Maybe. The only true chance you have of seeing a supercomputer in action is if you come up with a problem that will convince scientists to let you use some precious supercomputing time.

Where can I see a supercomputer?

If you ever find out, let us know too!

Actually, they're not that rare anymore. From processing financial transactions and



Not your desktop machine—the HP SuperDome HPC machine

simulating crash testing, to researching new cures in medicine, supercomputers are used the world over. Now all you need to do is get a job or break into one of the facilities that house one. Best of luck!